AI Based Recommendation System

Alian Khan

(03-3-1-007-2019)

Thesis submitted in partial fulfillment of requirements for the Degree of Bachelor of Sciences in Computer and Information Sciences

Department of Computer and Information Sciences,

Pakistan Institute of Engineering & Applied Sciences,

Nilore, Islamabad, Pakistan.

June, 2023

بسم الله الرحمن الرحيم

In the name of Allah, the Entirely

Merciful, the Especially Merciful

Department of Computer and Information Sciences,

Pakistan Institute of Engineering and Applied Sciences, Nilore,

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Approved by:

Signature:_____

Supervisor: ***Dr. Naeem Akhtar***

Verified by:

Signature:_____

Head, DCIS: Dr. Javaid Khurshid

Stamp:

This project is dedicated to my beloved family and friends, who offered me unconditional love and support throughout the course of this project.

Acknowledgement

First and foremost, I would like to thank Allah, our sole creator, helper, guider, endower of everything that we have in our lives and forgiver for all the sins that we are guilty of committing. If it wasn't for His enormous bounty, we would have never achieved anything in our lives and be lost.

I express my sheer appreciation to my supervisor Dr. Naeem Akhtar. He has always been keen on helping and has provided me the guidance on every aspect concerning my project. Even with his grueling schedule He has always taken out time to clear away our doubts or solved any problem which impeded my work.

Lastly but most importantly I would express my eternal gratitude towards my parents who have always been our backbone throughout the years and have never let me down no matter what.

Alian Khan

PIEAS, Nilore, Islamabad.

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Abstract

AI based recommendation system is an essential feature for various ecommerce platforms such as Ebay and Amazon to suggest desired products to end-users. Driven by the database and classical machine learning models, a variety of recommendation systems are available on these websites. However, conventional models are established based on machine learning techniques, which are unable to either provide accurate recommendations or take image-based inputs. The advancements in AI in general have paved new ways for development of recommendation systems. The aim of this project is to propose an AI based recommendation system. The starting point of the project is a state-of-the-art framework based on finetuning CLIP, which encompasses three phases: feature extraction and pre-processing, comparison with extracted embeddings and similar images recommendation phase. The framework is further improved by integrating it in a web application. Extensive experiments are conducted to evaluate the performance of the proposed model using real world data. By using a custom-made dataset, the newly trained model performs very well in both image and text domains respectively.

Chapter 1: Introduction

Electronic commerce, encompassing the promotion, servicing, production, delivery, sale, and payment for goods and services through electronic means, has become an integral part of nearly every business. Its global accessibility, user-friendly interface, diverse product offerings, secure payment techniques, and convenience of shopping from home have significantly enhanced the lives of consumers. Today, e-commerce applications enable users to access a wide range of services such as ticket purchase and sale, hotel reservations, food ordering, and the buying and selling of various products, including houses. Leading e-commerce platforms like Alibaba, Amazon, eBay, and Daraz leverage modern technologies such as cloud computing, big data, social networks, and mobile devices to provide personalized and exceptional user experiences.

Among the various domains of e-commerce, online shopping has gained substantial interest in recent times. It allows users to conveniently purchase goods and materials over the internet, eliminating the need to physically visit stores. Additionally, online shopping saves users valuable time by streamlining the process of searching for specific items that may be unavailable in certain physical stores. To further enhance the online shopping experience, businesses have turned to advanced technologies, particularly recommendation systems. Recommendation systems utilize user data and behavior to provide personalized recommendations for items, services, or products that are most likely to meet the user's preferences. In the realm of e-commerce, recommendation systems are extensively used across a wide range of services, including banking, restaurants, shopping, hotels, and real estate, to enhance customer experiences. By offering users additional relevant items that they may not have otherwise discovered, recommendation systems augment retail businesses. These systems employ a combination of classical methods and machine learning algorithms to generate personalized recommendations, with personalization being just one application among the myriad of services they offer.

While recommendation systems have gained popularity in e-commerce, traditional approaches like collaborative filtering and content-based filtering have inherent limitations. Collaborative filtering often lacks diversity in recommendations as it relies solely on past user interactions, potentially resulting in a narrow set of suggestions. Conversely, content-based filtering focuses solely on item attributes, disregarding user preferences. To overcome these limitations, deep learning techniques have been applied to recommender systems. Deep learning models can effectively analyze vast amounts of data, encompassing user behavior,

images, and text, thereby enabling more accurate and diverse recommendations.

In this thesis project, we propose an AI-based recommendation system for online shopping that utilizes both text and image inputs to generate similar images as outputs. Leveraging deep learning techniques, the system will analyze the input data and provide a set of related images based on the customer's preferences. The primary objective of this project is to develop a system that can deliver accurate and relevant recommendations to customers, thereby enhancing their shopping experience and driving sales for businesses.

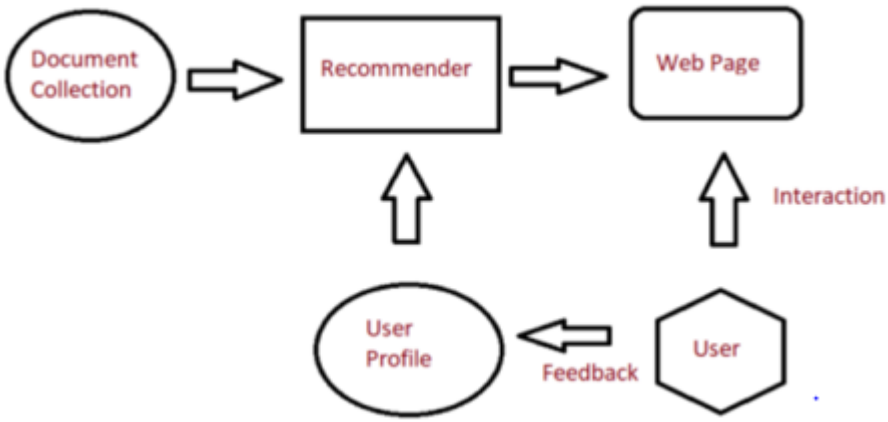

Figure 1. 1. Recommendation system workflow[1]

1.1 Problem Definition

Following the rapid boom of Ecommerce, Online shopping sites like Daraz, Amazon and Alibaba need to keep their users engaged and provide them with the services that they require. Thus, it has become an integral part of their policy to improve their recommendation systems to match them to the user’s requirements.

Although Recommendation systems being great for personalization have their own limitations. Sometimes these systems fail in to suggest relevant and desired items Thus disrupting the overall user experience. Modern recommendation systems are mainly facing the following two types of problems.

1.1.1 Irrelevant patterns

Recommendation systems suggest items based on similarities between 2 items. The greater the similarity, the more likely it will be for the user to find that item useful. Now modern-day recommendation systems have an issue of accuracy. These systems sometimes fail to suggest products which are desired by the user. This issue of irrelevant or unwanted suggestions greatly affects the user’s experience. This in turn impacts the overall sales of that store.

1.1.2 Undefined patterns

Present day Recommendation systems are a great help if we can define our input, but they are not good when they come across some items that cannot be described in words. For example, a customer wants to browse for some products whose shape could be defined in some known geometric patterns then he can easily feed in the details of that pattern into the websites recommendation system which in turn will suggest the similar available products. Contrary to this, if a customer browses a website for some product whose patterns do not lie into any known geometric pattern domain, then a problem arises of describing that pattern into words. As far as modern recommendation systems are concerned, they only deal with textual based inputs thus this problem lies far beyond their grasp.

1.2 Aims and Objectives

The main objective of this thesis project is to develop an AI-based recommendation system for online shopping that takes an image or text as input and provides related images as output. The system will use deep learning techniques to analyze the input data and generate a set of related images based on the customer's preferences. The system will be evaluated based on its accuracy, speed, and efficiency, and compared to traditional recommender systems.

To achieve this objective, the following specific objectives have been identified:

1. To identify the key features that customers look for when searching for products online.
2. To collect a large dataset of product images and text descriptions.
3. To develop an AI-based recommendation system using deep learning techniques such as Computer vision and natural language processing.
4. To evaluate the performance of the system using standard evaluation metrics.
5. To compare the performance of our system by evaluating it on real world text and image queries.

1.3 Motivation

The motivation for this project is to overcome the shortcomings of current e-commerce recommender systems, with the goal of giving customers a more enjoyable and gratifying buying experience. Businesses may increase customer satisfaction and boost revenue by using AI-based recommendation systems that provide personalized recommendations that closely match each client's particular tastes. These cutting-edge solutions utilize artificial intelligence to analyze large databases and give customized suggestions that are highly resonant with specific consumers. The goal is to provide a setting where clients feel really connected, supported, and inspired, leading to the development of long-lasting connections between businesses and their esteemed clientele.

1.4 Scope of project

The scope of this project includes designing an AI based recommendation system for online shopping which will take an image and text as input and will give similar images as output. This system will be developed using deep learning techniques and will be able to provide users with precise and accurate recommendations.

This proposed system will be trained on a data set consisting of images of different products and their textual descriptions which correctly identify the relationship between products and user preferences. This proposed system will take an image or text as input. It will extract features from these inputs. It will compare these features and give us the most similar images as output.

This project will aim to develop a system from the proposed model which will be evaluated using different metrics like accuracy and loss. This model will be deployed using a web application. This model will then be tested on real-world data to find its effectiveness. The application will be user friendly and easy to use.

1.5 Significance of study

This project proposed in this thesis can greatly impact the business of ecommerce stores by increasing sales and customer satisfaction. An AI-based recommendation system that works in both textual and image domains can boost customer engagement and activity by helping customers find products that match their preferences more easily, leading to increased customer satisfaction, activity, and loyalty. This system can also be very helpful for customers by saving their time and effort. Additionally, businesses can benefit from increased sales due to more accurate and diverse recommendations provided by the AI-based recommendation system.

1.6 Thesis organization

Chapter 1: Introduction

The objective of the project is made obvious in chapter one by the clarification of all the crucial facts. The issues and objectives that must be reached for the project to be completed are established depending on the existing system being employed.

Chapter 2: Related Concepts

The related concept's part contains all the related concepts to our project. The concepts that need to be studied before going through with the project.

Chapter 3: Literature review

The literature review is the main topic of chapter two. The need for the proposed project is explored in relation to the current system. The proposed idea must be superior to the current system.

Chapter 4: Methodology and implementation

In the Methodology part, all the project workflow is discussed. What needs to be done and how it tis going to be implemented. We then implement all of these in the implementation part.

Chapter 5: Results

In this chapter, we discuss all the results that have been achieved after system implementation. It is done through various graphs and comparisons. The system is tested in a real-world scenario.

Chapter 6: Conclusion and future work

The summary of the project is discussed in chapter 6. What has been done, what could be achieved, and what type of improvement could be made to the project.

Chapter 2: Data retrieval

2.1 Natural Language Processing

Natural language processing is a field of artificial intelligence that has gained much popularity recently. This field makes the computer learn how to understand, speak, and write human languages [2]. Natural language processing is not a new field, it has been around here for 50 years. This system has many applications in various fields like games, medicine, self-driving cars etc. NLP methods can learn any type of human language and in any form whether spoken or written due to its complex deep learning algorithms. These models can make computers learn and process all this information just in the same way as humans process the information that it receives through its various senses like hearing and speaking.

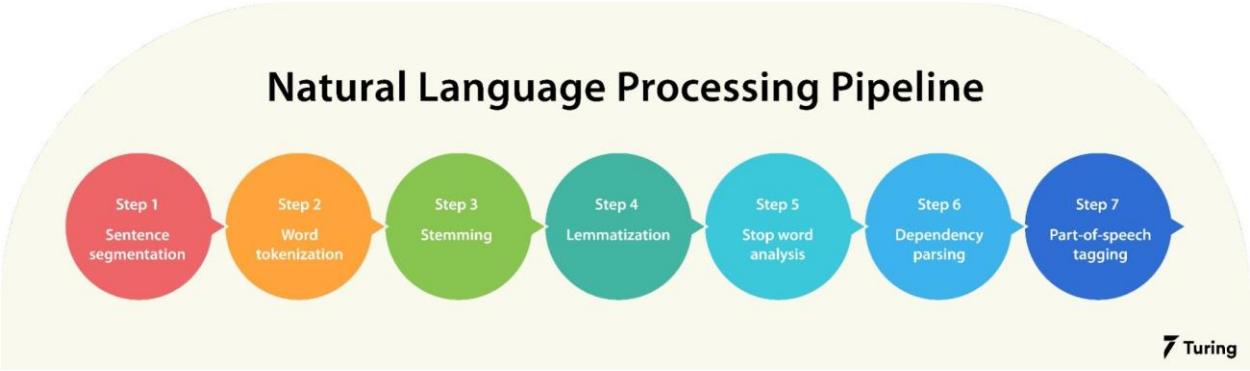

Figure 2. 1. Natural language processing pipeline[2]

2.2 Text to image retrieval

Text-to-image retrieval (also known as "text-based indexing") is the process of searching for images in a database based on the textual content associated with them[3]. This can be useful for tasks such as searching for images in a photo library or online image repository based on keywords or tags.

Text to image retrieval can be performed using variety of ways like metadata or using OCR (optical character recognition) to extract text from the images themselves, or some type of machine learning algorithm can be used for extraction of features from text and then comparison with images for finding most similar images.

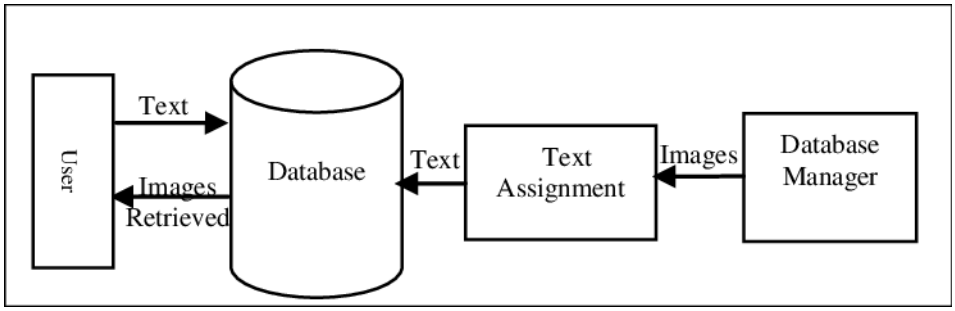

Figure 2. 2. Text to image retreival process[4]

2.3 Image to Image retrieval

Image to image retrieval means feeding an image to a system which will extract e feature s of that image and will compare to a dataset of another images. After comparison the system will then give us similar images as output. This system can be used in variety of tasks like recommendations, references and many more[5].

For the image-to-image retrieval process to be successful it is essential that we extract some type of information from the image. This information can be extracted in a variety of ways like feature extraction. After extracting the features, these extracted features can then be used for comparison with different images and retrieval of most similar images.

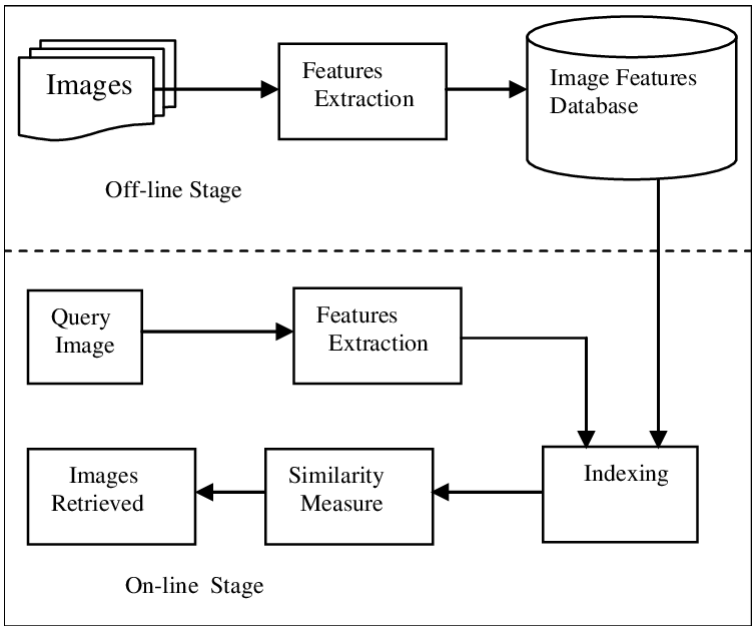

Figure 2. 3. Image to image retreival process[5]

2.4 Finetuning

A machine learning model that has already been trained can be fine-tuned to perform a new

job. Although maintaining the majority of the original model's parameters, it entails retraining the model using a fresh dataset. A model's performance on a new job can be enhanced by fine-tuning, particularly if the new work is linked to the original task and the new dataset is small. Consider the case when you already have an image classification model that has been trained on a sizable picture dataset. By retraining the model on the new dataset, you may fine-tune it to categorize a fresh batch of photos., setting the learning rate very low to avoid the model losing the features it picked up during the first training. The process of fine-tuning differs from training a model from start, in which all model parameters are randomly initialized, and the model is trained on a fresh dataset from scratch. In most cases, fine-tuning takes less time and data than starting from scratch when training a model.

2.5 Flask

Flask is a widely used software for python web development framework that allows developers to create web apps fast and effortlessly[6]. Flask is popular among developers who wish to build web apps rapidly because of its simplicity, flexibility, and ease of use[7].

Flask is a micro-framework, which implies it is a lightweight framework that just provides the minimal necessities for developing online applications. This makes it a good choice for small to medium-sized web apps since it helps developers to focus on implementing essential functionality without getting mired down in excessive complexity.

2.6 React

React is a front-end JavaScript open-source framework for building component-based user interfaces. [8]. It is used to quickly and efficiently construct interactive user interfaces and internet applications in comparison to utilizing plain JavaScript. Using React, you can create reusable components that you may think of as individual Lego pieces to develop your projects. When combined, these pieces form the whole user interface for the program. They are individual components of a final interface[9].

Chapter 3: Literature review

Recommender systems have become an essential part of ecommerce, providing personalized recommendations to customers to improve their shopping experience. There are many techniques used for recommender system development, two of which are content based filtering and collaborative filtering, but these methods also have many limitations like lack of accuracy and inability to work in both image and text domains.

Now due to the advancement of deep learning in almost every aspect of world, there have been developed some techniques which use deep learning for training of recommender systems. These systems improve the overall quality of recommender systems because these can learn very complex patterns and associations in data. This is possible due to training of these models on large-scale real-world datasets.

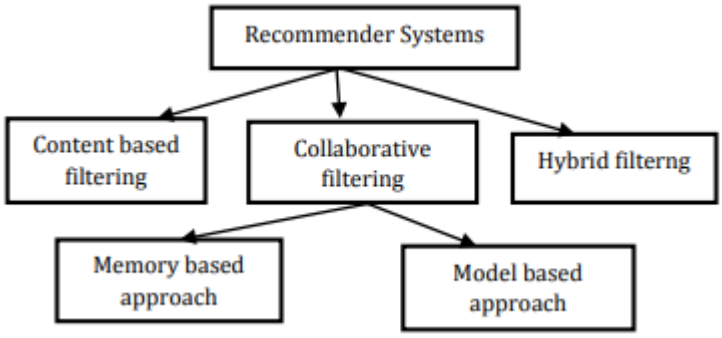

Figure 3. 1. Types of recommendation system[10]

3.1 Content based filtering

These types of recommender systems use machine learning techniques for recommendations given to the user. Only with a clear description of the product's features and a list of the user's options is it feasible to recommend items based on their attributes[11]. The recommender system builds a user profile using information about previous users' clicks, ratings, and likes. Future recommendations are more accurate the more a client interacts.

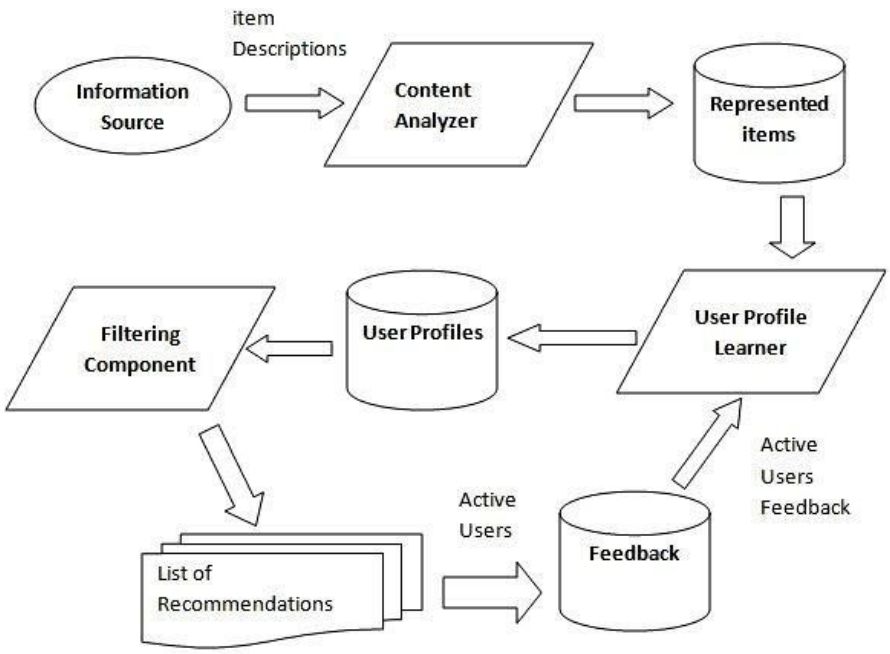

Figure 3. 2. Content based filtering architecture[12]

Deep learning and content-based features are used to create a cutting-edge music recommendation approach in the study "Deep Content-Based Music Recommendations" by Aäron van den Oord, Sander Dieleman, and Benjamin Schrauwen[13]. To extract pertinent properties, they train a CNN on spectrograms created from audio waveforms. Low-dimensional embeddings are produced by CNN for similarity-based recommendations. The accuracy and variety of the content-based suggestions are confirmed by a user poll, and experimental findings show that the content-based method is superior to collaborative filtering.

The research “Factorizing Personalized Markov Chains for Next-Basket Recommendation” suggests a customized Markov chain model for the suggestion of the next basket[14]. To provide personalized suggestions based on a user's current basket, it considers user preferences and item co-occurrence patterns. The model provides a workable solution for next-basket suggestion by outperforming other techniques in terms of accuracy on real-world datasets.

Using neural networks for recommendation systems, the paper "Neural Collaborative Filtering" suggests a unique method that blends collaborative filtering and content-based filtering[15]. The authors offer a neural network design that efficiently gathers information about item content and simulates user-item interactions. The model uses collaborative filtering signals and item content embeddings to produce personalized suggestions.

According to experimental findings, this strategy performs better than more established ones, offering recommendations with greater accuracy and diversity.

Convolutional matrix factorization (CMF) is a revolutionary technique that includes document context, as described in the paper "Convolutional Matrix Factorization for Document Context-Aware Recommendation"[16]. The method takes advantage of contextual data for recommendation tasks and successfully captures the sequential character of document data. The CMF model learns latent factors that mix content-based and contextual information by using convolutional operations on document matrices. Experimental tests support the idea that by taking document context into account, suggestion accuracy can be increased.

3.2 Collaborative filtering

Collaborative filtering-based recommender systems only depend upon past interactions between users and items to suggest new products. All item's unique characteristics are not considered. The historical data about the user's interactions with the objects is saved and preserved throughout collaborative filtering. User-item interaction matrixes are comm Collaborative filtering-based recommender systems solely rely on past interactions between users and items to suggest new products. All item's unique characteristics are not considered.

The historical data about the user's interactions with the objects is saved and preserved throughout collaborative filtering. User-item interaction matrixes are commonly used to depict this, with rows denoting users and columns denoting items. When creating suggestions to the target user, similar people are grouped together, and all their interactions are considered. only used to depict this, with rows denoting users and columns denoting items. When creating suggestions to the target user, similar people are grouped together, and all their interactions are considered.

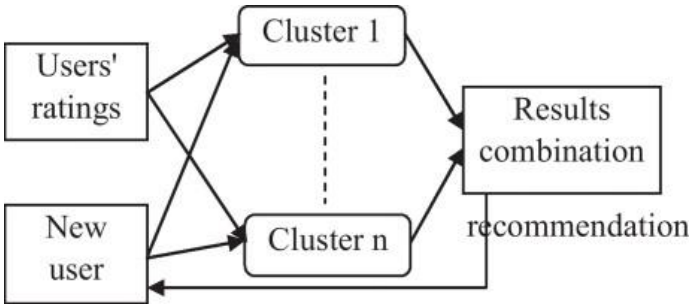

Figure 3. 3. Collaborative filtering architecture

Ruslan Salakhutdinov and Andriy Mnih's proposed a new method in their research

"Bayesian Probabilistic Matrix Factorization for Collaborative Filtering"[17]. This new technique was using a Bayesian probabilistic matrix factorization technique for collaborative filtering. The method models uncertainty using Bayesian inference, which is particularly helpful in circumstances with sparse data and cold-start issues. Bayesian concepts are included in the model to produce more reliable and accurate suggestions. Experimental findings show that the suggested strategy is successful in improving suggestion quality.

In this research “CFM: Convolutional Factorization Machines for Context-Aware Recommendation” by Xin Xin, Bo Chen, Xiangnan He, Dong Wang, Yue Ding and Joemon M. Jose[16]. The authors provide a unique method for collaborative filtering that makes use of a deep learning framework and a marginalized denoising autoencoder. By accurately recording user and object interactions, the technique seeks to acquire latent representations of users and items. The model creates customized suggestions by fusing collaborative filtering signals with item content embeddings. Experimental findings show that the deep collaborative filtering strategy performs better than conventional techniques, offering more precise and varied recommendations.

Another research conducted by Yao Wu, Christopher DuBois, and Alice X. Zheng's called "Collaborative Filtering with Stacked Denoising Autoencoders and Sparse Inputs"[18]:

The collaborative filtering technique described in this research successfully handles sparse inputs and involves stacked denoising autoencoders. The method is intended to address the difficulties presented by high-dimensional, sparse data frequently seen in recommendation systems. The model captures complicated user-item interactions by learning latent representations of people and objects, which leads to better recommendation generation. Experimental analyses show that the suggested approach is beneficial in terms of suggested accuracy.

3.3 Hybrid filtering

The hybrid recommendation system concurrently leverages data from collaborative data and content-based data to present customers with a comparable or nearby item. It is sometimes simpler to handle the bigger challenges when the two approaches are combined. Here, the algorithm suggests products that are comparable to those the user already uses or those that other users who have specific traits with the user are likely to use.

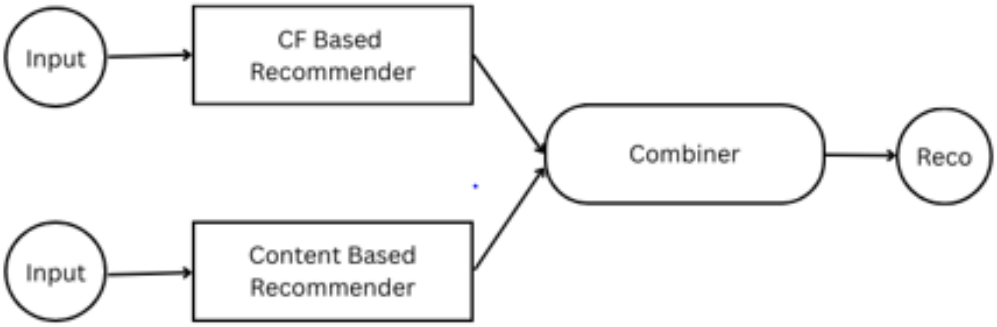

Figure 3. 4. Hybrid filtering architecture

In "Hybrid Recommender Systems: State of the Art, Challenges, and Opportunities," Ricci, Rokach, and Shapira give a thorough overview of hybrid recommender systems [9]. This study looks at many methods for fusing content-based filtering, collaborative filtering, and other recommendation strategies. It explores the difficulties involved in creating and putting into use hybrid systems while highlighting the advantages they provide. This book may be used as a thorough reference for learning the area and offers insightful observations on the present state of the art in hybrid recommender systems.

Burke performs a comprehensive analysis of hybrid recommender systems in his article "Hybrid Recommender Systems: A Systematic Review"[19]**.** The merging of several recommendation techniques, including collaborative filtering, content-based filtering, and knowledge-based filtering, is the main topic of the study. Burke looks at several integration strategies for these methods and offers a thorough analysis of the advantages and drawbacks of using hybrid approaches in recommendation systems. Understanding the state-of-the-art in hybrid recommender systems and their consequences is made possible by this review.

Su and Khoshgoftaar give a thorough analysis of hybrid collaborative filtering methods for recommender systems in this study “A Survey of Collaborative Filtering Techniques.”

[20]. The authors go over several ways to combine collaborative filtering with other types of recommendation techniques, including content-based filtering, demographic filtering, and knowledge-based filtering. They emphasize the elements that affect hybrid approaches' success as they evaluate their benefits and drawbacks. The study presents suggestions for future research in this area and offers insights into the current state of the art in hybrid collaborative filtering.

The following table summarizes all the content of all these methods suggested in various research discussed above.

Table 1. Comparison of research papers

	Collaborative filtering	**Content based filtering**	**Hybrid filtering**
Techniques	.TF-IDF . Clustering . Naïve bayes . Probability	. KNN . SVD . Clustering . Probability . Tf-idf . Neural Networks . Apriori algorithm . Vector distance algorithm	. Clustering . KNN . TF-IDF . Matrix factorization . Probability
Advantages	It works when there are no reviews on a product.	It offers a serendipitous suggestion for the consumers, therefore there is no need to take the content of the recommended things into consideration like with content-based filtering. It is simple to deploy, relies little on data, and produces reliable	Overcome the challenges with the cold start, data sparsity, and overspecialization

		recommendation results.	
Disadvantages	Overspecialization makes it challenging to deploy on vast product databases since users have varying viewpoints on each item.	When dealing with enormous volumes of data, there is poor scalability. A chilly start. Difficulty with data sparsity. Scalability	The cost of implementation is high. Hybridization increases the complexity of the recommendation process.

In most of research conducted, the common point between all models is that most of them are based on either only image to image-based recommendation systems or image to text-based recommendation systems, but in today's world the need for a multimodal recommendation system is too much necessary due to the complexity of designs in clothes. This method proposes a solution for this problem which is based on a hybrid model. The author used a model developed by Open AI called CLIP for the development of multimodal recommendation system. This resulted in better overall performance of individual queries of both text and images and resolved the issues.

3.4 CLIP

CLIP (Contrastive language image pre-training) is a training method introduced by OpenAI for zero shot classification of image-text pairs. It is a revolutionary model which when compared with ResNet 50 on the original ImageNet dataset gives result on same level as the algorithm but with zero shot classification i.e. without training the model on original 1.28M labels[21]. CLIP features outdo ImageNet best 50 features.

3.4.1 Architecture and training

Clip has been trained on a dataset called LIASON which consists of 400M text and image pairs. CLIP specializes in zero shot training and classification. Clip training consists of 3 major parts[22].

3.4.1.1 Contrastive Pre-training

CLIP uses 2 encoders, one for text and one for images. Both produce a 512-size embedding thus creating a symmetric matrix of size 512*512.

3.4.1.2 Create dataset classifier from label text

In this step, CLIP takes of its text classifiers and attaches a sentence to them. It then tokenizes them and is passed through a text encoder where the embeddings are produced.

3.4.1.3 Zero-shot prediction

In this step, we pass our image through an image encoder, and it gives us image embedding which is then compared to all text embeddings and through cosine similarity index we find the matching texts to image.

3.4.2 Text to image retrieval

Clip uses symmetric matrix of text and image as discussed in section 5.1.1. This symmetric matrix can be used to translate an image embedding into a text embedding and vice versa.

3.4.3 Image to image translation

As clip uses symmetric matrix and produces a 512-size embedding thus it can be used to translate one image to another image. Like text image to text classification discussed in section 5.1.3, it can be used to do image to image classification based on cosine similarity index.

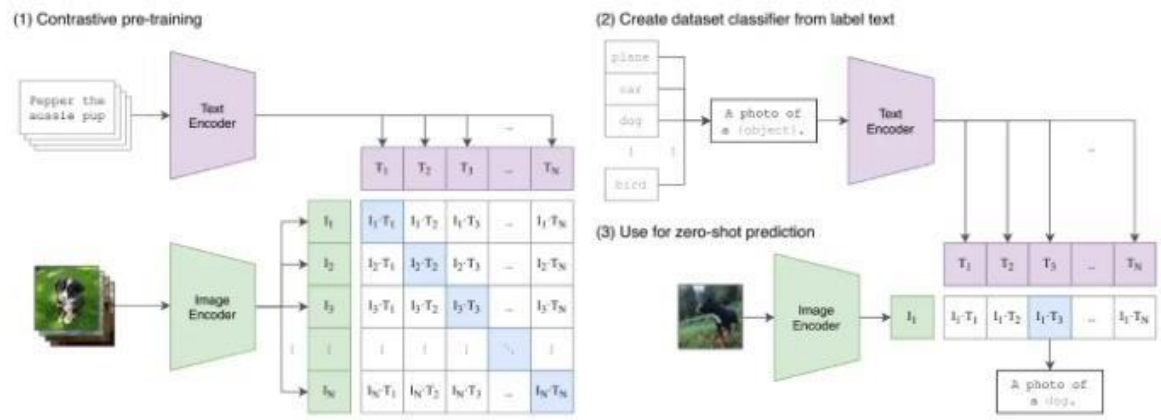

Figure 3. 5. CLIP working[23]

Chapter 4: Methodology & Implementation

4.1 Overview of project workplan

Fig4.1 shows the general workplan for our project. First, A dataset will be built consisting of different types and shapes of clothes patterns. Each of these images of cloth patterns will be associated with a text describing the image accurately. A pretrained model like OpenAI CLIP will be selected which will extract features from our data. For finetuning of the model, the dataset will be divided into training and testing. This data will be fed into a pre-trained model. The hyperparameters will be adjusted accordingly during training and the model with the best results will be selected. A web application will be built for this model. This web application will be deployed on to an online server. The details of all this process will be discussed in the upcoming chapters.

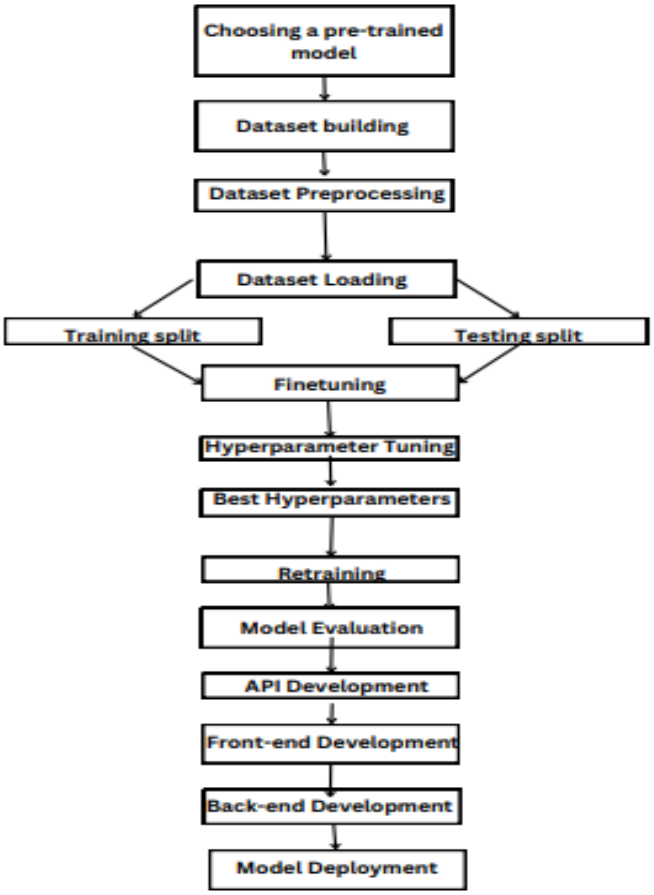

Figure 4. 1. Project wrokplan

4.2 Proposed model framework

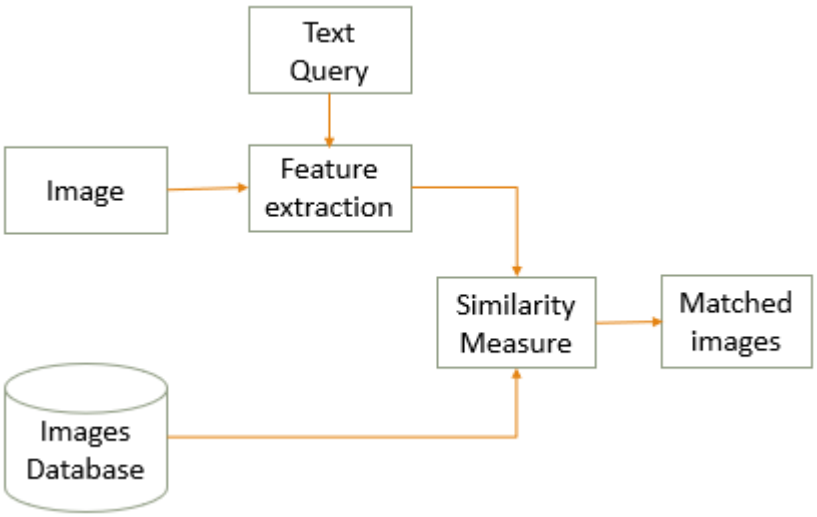

Figure 4. 2. Model framework

The model shown in fig4.2 is the proposed model for the problem discussed in chapter1. This model is based on Natural language processing and computer vision techniques. It uses both techniques because of the specific accuracy and problem requirements. Considering these things, this proposed model includes a pre-trained machine learning model which will take an image or a text as input. This model will perform classification on these inputs. The algorithm used in this process will be KNN. We use KNN algorithm because it can easily tell us similar items in a feature space. In this way we can efficiently find similar images.

The model will consist of 2 separate encoders for both image and text. The image encoder will take image as an input and give us an embedding vector as output. While the text encoder will also follow the same principle. As the model will be multimodal, it should make the same size embeddings for text and image.

After the embeddings have been set the model will then assign all these embeddings the same vector space. In this space similar images will be found using cosine similarity measures. The vectors having greater cosine similarity will lie closer and the ones having dissimilar cosine similarity score will lie far away.

4.3 Model selection

This section will cover the process of choosing a model for a multimodal translation model that can translate text to image and picture to image. Finding a model that not only facilitates multimodal translation but also exhibits high accuracy in generating trustworthy and high-

quality translations was the main goal of the selection process. After extensive investigation and review, OpenAI's CLIP model came out to be the best choice for the assignment.

4.3.1 Research and Review

To investigate current multimodal translation models, a thorough literature analysis was done.

Models that can translate from text to picture as well as from image to image were found and given consideration for review. Important aspects were evaluated, including model design, translation task performance, and handling of multimodal inputs.

4.3.2 Evaluation criteria

Appropriate assessment measures were used to guarantee the precision and caliber of translations. Metrics like RMSE and image and text accuracy were taken into consideration for text-to-image translation to gauge how comparable and coherent the output visuals were to the source text. Same measures were selected to measure the visual quality of the retrieved pictures for image-to-image retrieval.

4.3.3 Selection of OpenAI CLIP

OpenAI's CLIP model was found to be the best option after rigorous deliberation and comparison for the multimodal retrieval job. In both text-to-image and image-to-image retrieval challenges, CLIP excelled, displaying its adaptability and precision. The design of the model, which makes use of contrastive learning and pretraining on huge datasets, helps it handle multimodal inputs well.

We chose OpenAI's CLIP model because it combines accuracy and resilience to produce high-quality translations in both the text-to-image and image-to-image domains, providing a solid basis for our multimodal translation effort.

4.4 Environment setting

4.4.1 Linux

For this project, A dedicated PC was used with an ubuntu 20.4 installed in it. This operating system was used to perform training and evaluation tasks for our project. All our intended libraries and models were installed in this system.

4.4.2 Pytorch

Pytorch is a machine learning framework whose architecture is based on the torch library of python. This library can be used to perform various kinds of tasks like natural language processing, computer vison and different deep learning techniques. This framework was used in our project for finetuning of our pretrained model.

4.4.3 CUDA

CUDA is a library for GPU programming. It is used to perform various GPU based tasks like loading graphics, converting sequential processing into parallel processing and many more. In our project we use CUDA for parallel preprocessing of our dataset and training of our model.

4.4.4 CLIP

CLIP is a model developed by OpenAI. It is used for various tasks like text to image classification, image to text classification, visual question answering, and image to image classification. In our project, CLIP is used as the pretrained model which is to be finetuned using transfer learning.

4.4.5 Python

Python is a programming language vastly used in the world for machine learning and data science. It contains many libraries for all types of tasks like object detection, image detection, data loading, data cleaning, game development, app development and data analytics. In our project we use Python 3.9 because it is compatible with all the latest libraires used in our project.

4.5 Dataset Building

This section describes the strategies used to create the dataset, which included using the Sele nium library and web scraping methods. For our project we required a dataset which would consist of different types of images for various clothing patterns of different categories. This dataset would also contain the textual descriptions for all these images. There was no publicly available dataset which would fulfill this requirement so to progress further it was a requirement for us to make our own dataset. We built our own dataset using web scraping tools and methods. This dataset was composed of 30832 images. All these images were associated with their textual descriptions. These images varied in shape, size, and resolution.

The text associated with the images also varied in length. The text was comprised of maximum 77 context length.

Utilizing the sophisticated features provided by the Selenium library, a reliable web scraping technique was used to get the dataset. Selenium has established itself as the top option for effectively gathering data from dynamic web sites because of its adaptability and flexibility in automating web browser interactions. The demand to compile a sizable and varied collection of photos, accompanied by their accompanying verbal explanations, taken from a variety of web sources, served as the driving force behind this choice. The process of acquiring datasets was expedited and improved by utilizing Selenium's extensive toolkit. Selenium enabled the systematic extraction of photos and related text material from the specified websites by automating complex browser processes. To gather a complete and representative dataset, it was crucial that the tool be able to move across intricate web architectures, interact with dynamic elements, and record data in real-time.

A script was written in python which would navigate through the browser to specific website. The script first loaded all the images in the same category and then opened the images 1 by1.

After loading the URL’s, the script would then take an image, download it, and rename the image by numbering it sequentially. It then copied the details attached to the image and placed them in a CSV file. This CSV file was also generated by the script. CSV comprise of 2 columns, 1 for the name of all images and other for the details associated with that specific image. This process would repeat for a single category till the last image. After that the script would move on towards the next category and start loading all the images and then downloading them 1 by 1. As there were pictures that were repeated in multiple categories, a strict check was applied on the URLs of the images. This check would make a list of all URLs accessed so that no URL would repeat.

During the dataset collecting phase, great care was taken to respect ethical principles and legal compliance. The robots.txt and terms of service of the targeted websites were followed during the scraping procedure. Requests were made in a responsible manner, minimizing traffic and its effect on the effectiveness of the websites. Additionally, the privacy and intellectual property rights of the website owners and their users were protected by handling personal information and copyrighted content with the highest care and consideration.

4.6 Dataset Splitting

The dataset needed to be divided into a training set, validation set, and a testing set after it had been properly constructed. This division was done to make it possible to test and validate the built models on hypothetical data. The split ratio used in this study was 70:15:15, with 70% of the data set aside for training, 15% set aside for validation and 30% set aside for testing.

The choice of a 70:15:15 split was made to guarantee that there was a sizable enough training set to successfully train the models while still leaving a sizable amount of data for assessment. This division finds a compromise between maintaining a representative sample for objective performance evaluation and offering sufficient training data for model optimization.

While maintaining the class distribution in all the training, validation and testing sets, the datasets were split at random. By ensuring that each category is fairly represented in both subsets, this method avoids bias that may result from unbalanced class distributions.

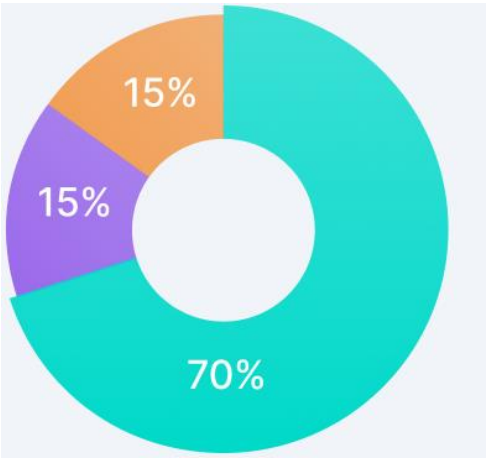

Figure 4. 3. Dataset split

Table 2. Dataset composition and split

Dataset	**Images**	**Descriptions**
Train	21500	21500
Validation	4625	4625
Test	4625	4625

Total	30832	30832

4.7 Dataset Preprocessing

In the case of our model as we are using the technique of transfer learning, so we don't need to define any explicit function for preprocessing, but the step is done by the model CLIP. When data is passed to CLIP then as we discussed earlier that all the images are of various sizes and various lengths. Now to train on these images and texts, CLIP must perform some steps to make all the images of the same size. The preprocessing steps that are used by CLIP are as follows.

4.7.1 Image preprocessing

Resize: As discussed earlier tha CLIP input consists of images with various sizes so to make them consistent, CLIP uses a preprocessing technique of resizing in which all the images used in our model are resized to a size of 224.

Center crop: Clip uses another preprocessing technique to make the images consistent which is called center crop. In this step clip crops all the images to same resolution and same pixel values of 224 * 224.

RGB: The input of CLIP contains images of both RGB, greyscale and black and white format. To make them consistent, we convert all the images to RGB format.

Normalize: We scale all our images within a range of 0 and 1. We set a standard mean and standard deviation.

4.7.2 Text preprocessing

A tokenizer is used to prepare the textual input for the CLIP model. This tokenizer handles text consistently regardless of capitalization by operating in a case-insensitive way. The text query is sent into the tokenizer, which creates a lengthy list of 77 tokens that are padded to a certain length.

The preprocessing pipeline has 72 characters defined as the maximum context length for the text. This suggests that a sentence may have up to 72 words. 72 of the padding list's 77 tokens are designated to stand in for particular words or word groups in the text. Within the tokenizer, the remaining five tokens are used for specific functions. Three of these tokens are used to denote certain instructions or components, giving the model processing context

information. To accurately identify sentence boundaries, the final two tokens are used to indicate a sentence's beginning and conclusion.

The CLIP model guarantees that the textual input is suitably prepared for future analysis by integrating this preprocessing phase. The model can efficiently handle and comprehend the input within the context of its job thanks to the tokenizer's conversion of the text query into a standardized token representation.

Table 3. Data prerprocessing techniques applied

Image	**Text**
Resize	Tokenization
Center crop	Padding
Normalize	Context length
Greyscale to RGB	Special tokens

4.8 Model training

After splitting the dataset and preprocessing, we move on towards the model training process. In model training we first convert our data from CSV into lists. These lists are then passed to a function which performs feature extraction on all the list entries. Each of these entries are then converted into a feature vector of size 512. Images and text are passed into separate encoders for extraction of embeddings. These embedding vectors are concatenated, so we get 2 lists of embedding vectors. One of these vectors is for images and 1 is for text.

Before the training process begins, we first set the parameters and hyperparameters for training. We use the optimizer Adam for training. The number of epochs set for training is 100. We use L1 and L2 regularization techniques to prevent the model from overfitting. We also use the momentum variable to prevent the model from stopping into a local minima and to improve the overall training process. An epsilon value of 1*10^-6 is used for smooth training. We use a learning rate of 1 * 10^-7 for the training of our model. The algorithm used for backtracking is gradient descent. This whole training process is carried out using Pytorch library.

We use transfer learning techniques for our training. First, we load our model CLIP which is

then called into the training process. The version of CLIP used is VIT-32 because of its greater number of attention heads. The training process used here is mixed precision training for CLIP. The model is loaded with GPU so the training process can be performed in parallel which decreases the training time significantly and increases the overall performance.

During training we save our model, parameters, and optimizer state. This process is set to 5 epoch interval meaning model is saved after every 5th epoch.

4.9 Web-app Development

The web-app for our model is based on the architecture in Fig4.3. This app will take an image or text input and forward it to the API. This API converts the image to Base64 format and sends it to ML model. ML model processes this input and returns 5 similar images which are sent back to web-app for display.

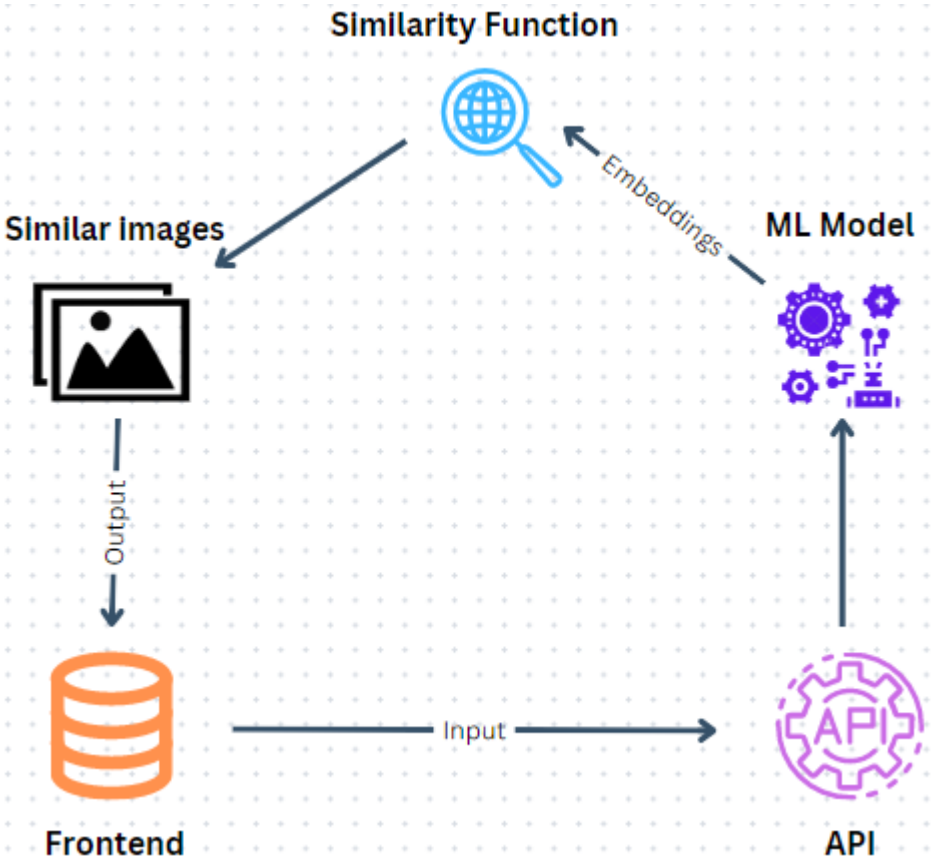

Figure 4. 4. API architecture

The Web-application development cycle is explained below.

4.9.1 API Development

Our Flask and Python-based API has two routes that may process input in the form of either text or pictures. The following actions are taken by the API:

4.9.1.1 Input Conversion

The API converts all received text and images into their matching base64 representations. This guarantees that the data can be processed and transferred without difficulty.

4.9.1.2 Passing Data to ML Model

The machine learning model is then given the converted input (in base64 format) to process. Based on the recognized patterns and correlations, the model searches for related pictures using the extracted embeddings from the input data.

4.9.1.3 Generating Output

Based on the input data, the ML model creates a collection of related photos. Then, to ease transfer and ensure API compliance, these related photos are transformed into base64 format.

4.9.1.4 API response

As a last step, the API delivers a collection of related photos. The pictures are in base64 format, making it simple for the receiving program or user to decode and display them.

4.9.1.5 API Testing

API is tested with Postman. A post request is generated and sent to the API. The API processes the input and returns similar pictures as output in Base64 format.

4.9.2 Front-end development

React was used to create the API's frontend, which has a modern and simple user experience. It has a single screen and emphasizes use and simplicity. To make user interaction and input selection easier, the interface includes a button and a selection menu. This simplified layout improves user experience and encourages effective use of the API's features.

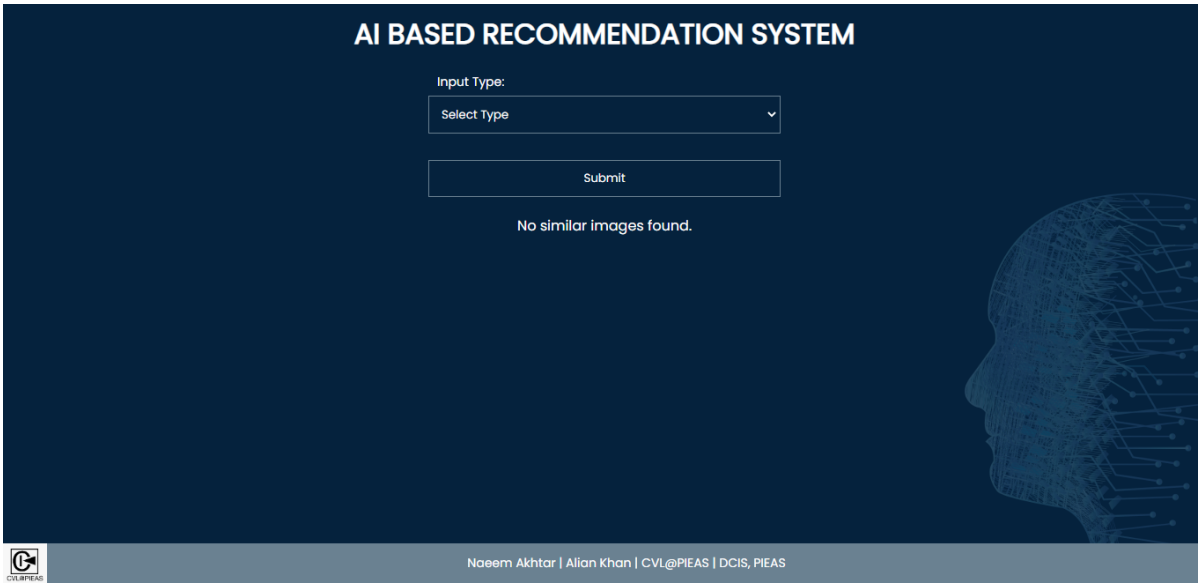

Figure 4. 5. Homepage

There is a dropdown menu with the options "Image" and "Text" on the user interface.

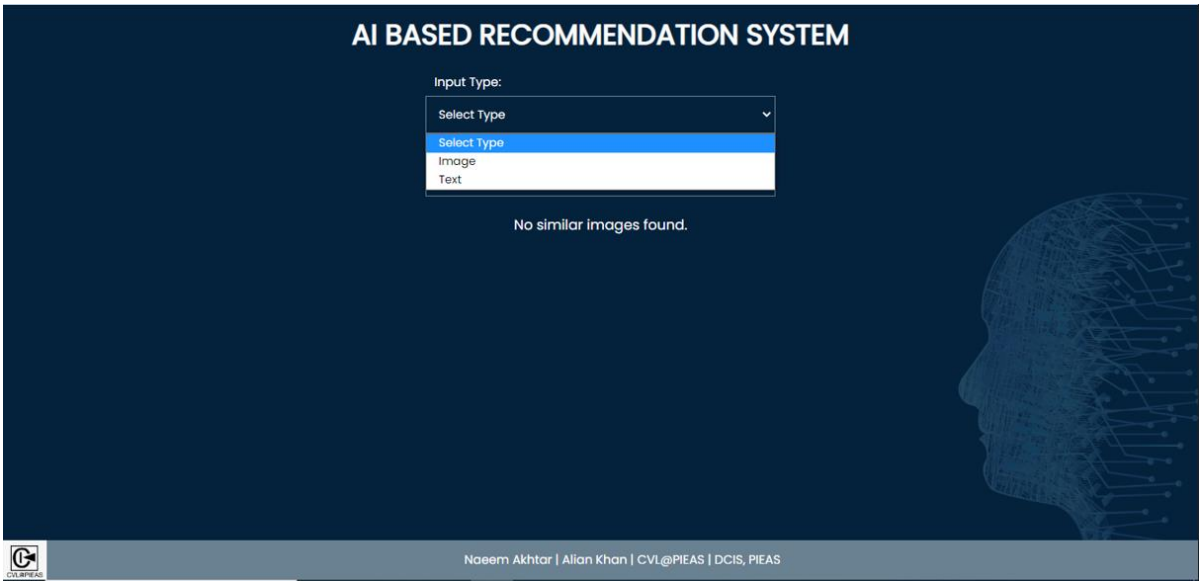

Figure 4. 6. Input selection screen

When the user chooses the “Image” option, a new button is produced on the fly. This button activates a browse menu that lets the user choose an image file from their local computer.

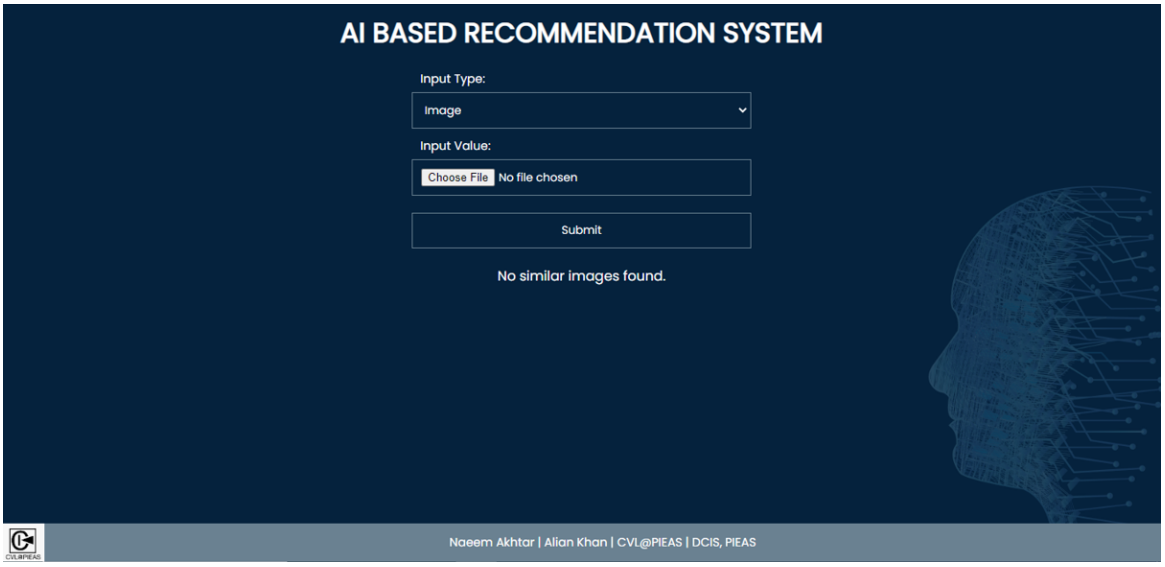

Figure 4. 7. Image input screen

A text input box appears instead if the user selects the "Text" option from the dropdown menu. With a word limit of 72 characters, the user of this text input box is free to insert any content they like.

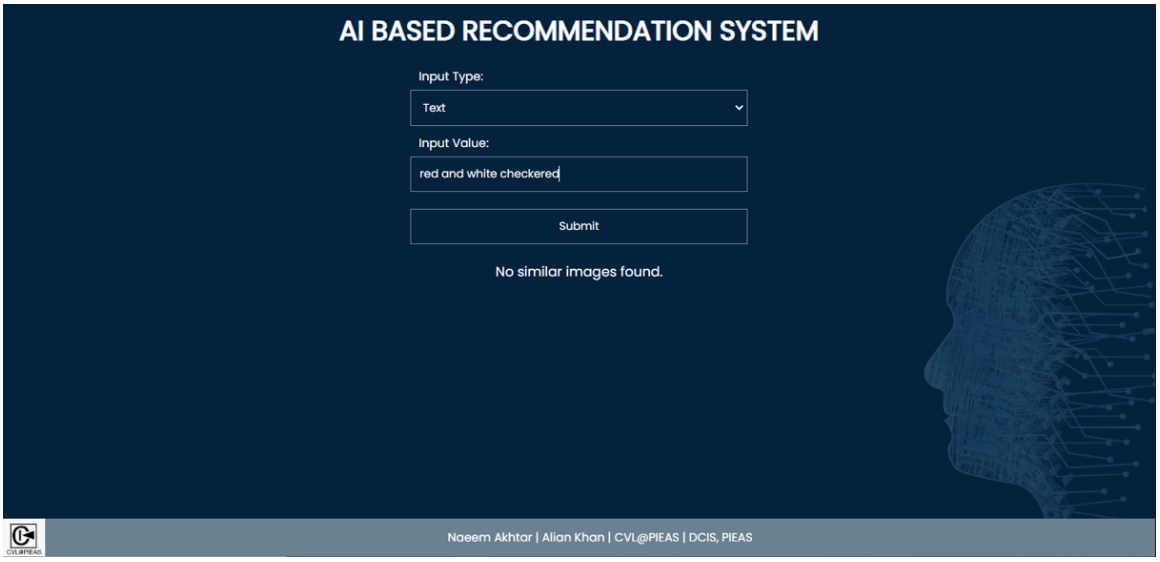

Figure 4. 8. Text query screen

The user then clicks the "Submit" button after typing the text or uploading the image. The input given is subsequently sent to an API for additional processing. This parameter is used by the API to find photos that are comparable based on predefined criteria. Finally, the API-retrieved identical photos are presented on the screen so that the user may see them and respond to them as necessary.

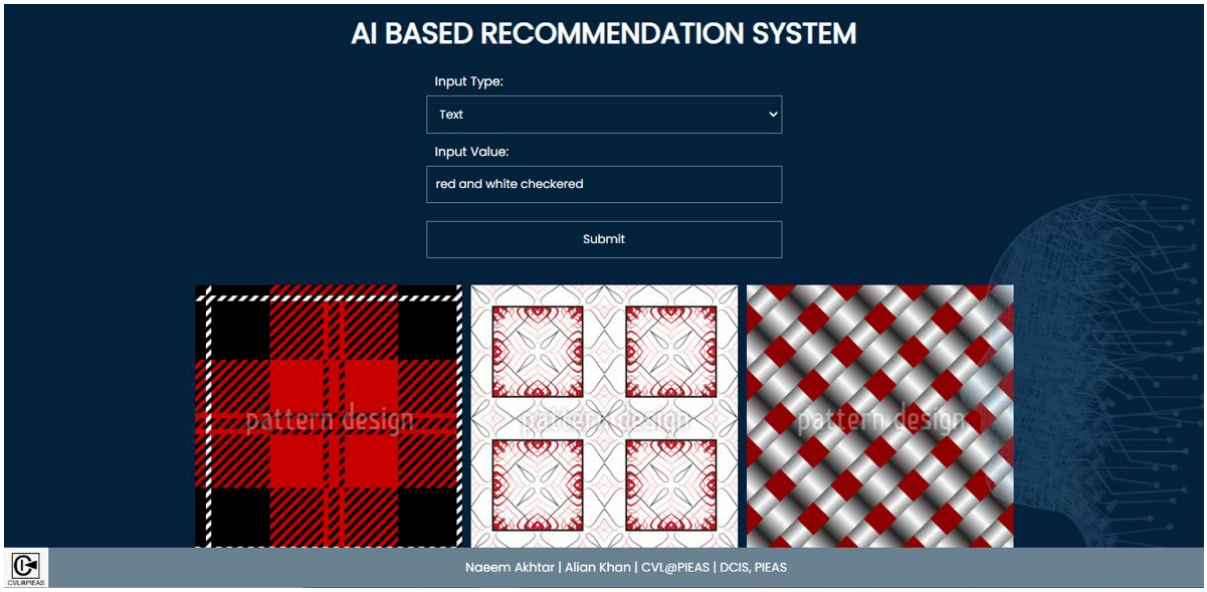

Figure 4. 9. Result screen

Chapter 5: Results and Discussion

Several experiments were conducted to assess the deep learning model framework suggested in this research. The accuracy values for both image and text encoders will be utilized as assessment measures while the deep learning model executes regression tasks. Additionally, RMSE is considered when evaluating. Measures including hyper-parameter tweaking, and its overall predictive performance are assessed. The following sub-chapters cover the specific findings from the various situations.

5.1 Training results

5.1.1 Hyperparameter Tuning

The ideal parameter for the proposed AI based recommendation system was determined using hyper-parameter analysis. Additionally, it can demonstrate the overall efficacy of the suggested structure. In this project analysis of the training ratio and learning rate was done. The analysis was done to find the ideal hyper-parameter.

The performance measurement of the proposed model at various learning rates was done. Training was carried out using learning rates of 5e-8, and 5e-7. The learning rate with the lowest RMSE, 1.07e-07, was attained at the 5e-7 learning rate. Since 5e-7 learning rate achieved the lowest RMSE, it is used as the default learning rate in the project.

The training ratio is the second hyper-parameter that was examined. In this section, the custom dataset's 70% and 90% were utilized as training sets, while the remaining 15% and 15% respectively was used as the validation and test set. The lowest RMSE value was achieved using 70% training set. However, it should be noted that the size of the training set was increased from 4500 to 21500.

The findings show that when the size of the training set is increased, the performance of the deep learning model continuously advances from a 4.42e-05 to a 1.07e-07 RMSE. When the dataset was increased from 5000 instances to 30833 instances the deep learning model has the lowest RMSE. This demonstrates how the deep-learning-based approach scales well with increased data.

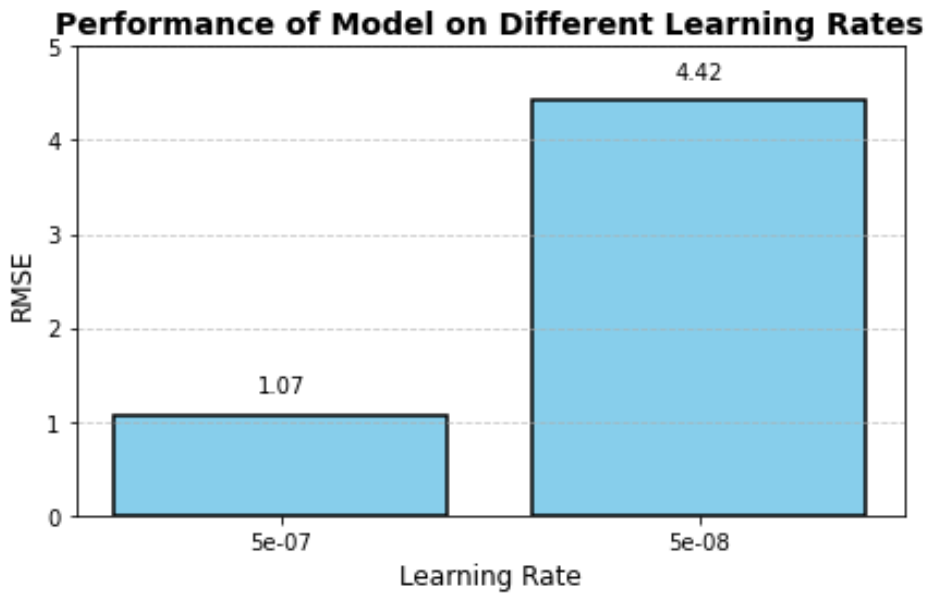

Figure 5. 1. Training result comparison

5.1.2 Graphs and results

The model was trained with 2 evaluation metrics: Accuracy, and loss. The graphs for all these metrics were generated at various time stamps in our training process. The results are given below.

At 100 epochs: The model was saved at 100 epochs the values of accuracy and loss were 99.92558139534884, and 1.005489324456328e-07. The validation accuracy at this timestamp was 80.44. Validation loss at this stage was 5.248380120053249e-05. The graphs generated at this timestamp are as follows.

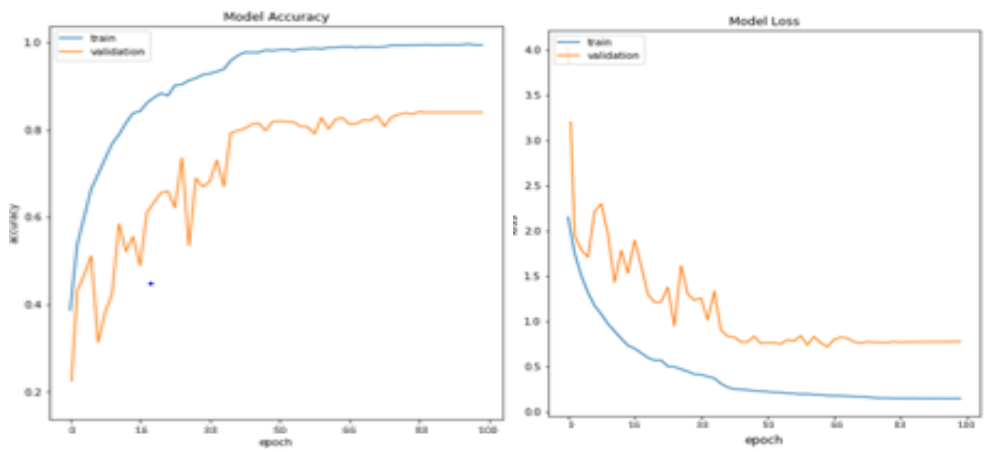

Figure 5. 2. Training and validation graphs

From the results it can be observed that we are getting a very high training score and an optimal validation score which shows that no overfitting of the model has occurred.

5.2 Evaluation results

For evaluation of the model, we prepared an evaluation dataset. This dataset was at first 10% of the data and second time we increased it to 30% of the data. For the first model we achieved an evaluation accuracy of 60%. But when we increased the overall data and the training data then we achieved results of 80% accuracy. These accuracies were achieved on the models with best training results as discussed in section 5.2.2. This showed a significant increase in accuracy by adding more data. This result comparison is shown in the bar graph below.

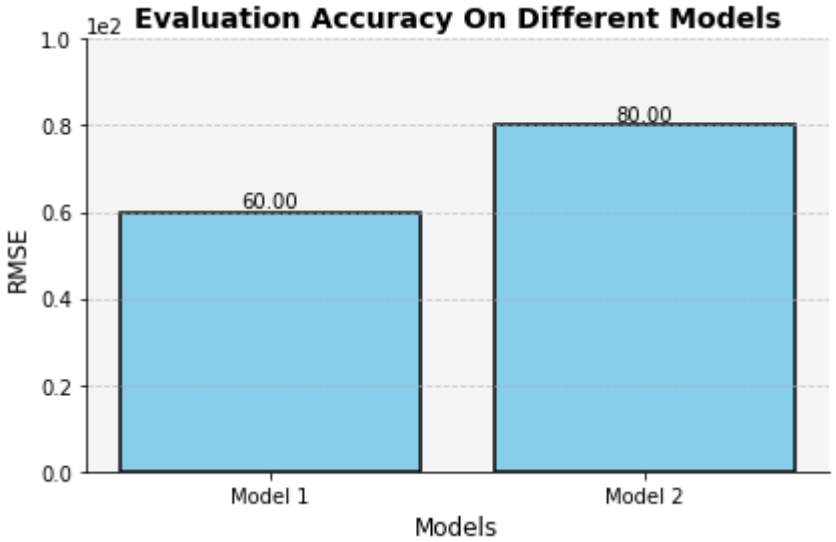

Figure 5. 3. Evaluation results comparison

5.3 Summary of results

It can be seen from the discussion above, the accuracy of our model varies greatly by varying training data size, data split and learning rate. When the overall data is increased the performance of the model generally goes up. This significantly improves the overall performance of the model. The learning rate variation and number of epochs for training also affects the accuracy of the model. A too high learning rate leads to overfitting while too little leads to underfitting as in the case of our model. The same goes for several epochs.

Chapter 6: Conclusion and future work

6.1 Conclusion

This project identified various problems that exist in traditional recommendation systems based on the literature review done. These problems include the issues that come with the traditional algorithms that these models were trained on, like not being able to work on in both text and image domains. This project was able to achieve all solutions to these problems as discussed in aims and objectives.

This project implemented an AI based recommendation system by finetuning CLIP model with a custom dataset obtained through web scrapping. This finetuning was done using Pytorch. The hyperparameters for this model were analyzed to obtain the best parameters for training. A web application for this model was built using Flask and react. The results of the model are evaluated on real world examples which shows significant results I both image and text domains.

In summary the proposed model “AI based recommendation system” can predict accurately various clothes patterns more efficiently than most of the recommendation systems. It is also capable of taking both image and text-based inputs.

6.2 Future work

This final year project also suggests that for future progress following area are worth to be explored in the future:

It is suggested that the accuracy of the model can be further increased if the quality of the dataset is improved, and the overall quantity of dataset is increased. If an even more efficient training process is developed using much more tuned hyperparameters the accuracy can be further increased.

Finally, it is proposed that the AI based recommendation system be integrated into a real-world fashion application or website to assess how beneficial it is for recommending clothes in real-time.

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